MIT 6.1200J
Problem Set 5

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Problem 1: Fibonacci Divisibility

(a) **Claim:** For every $n \geq 0$, $F_{n+a} \equiv F_n \cdot F_{a+1} \pmod{F_a}$.

Proof by strong induction on n.

Base cases.

- $n = 0$: $F_{0+a} = F_a \equiv 0 = F_0 \cdot F_{a+1} \pmod{F_a}$. ✓
- $n = 1$: $F_{1+a} = F_{a+1} \equiv F_{a+1} = F_1 \cdot F_{a+1} \pmod{F_a}$. ✓

Inductive step. Let $n \geq 2$ and assume the statement holds for $n-1$ and $n-2$. Then:

$$F_{n+a} = F_{(n-1)+a} + F_{(n-2)+a} \equiv F_{n-1} \cdot F_{a+1} + F_{n-2} \cdot F_{a+1} = (F_{n-1} + F_{n-2}) \cdot F_{a+1} = F_n \cdot F_{a+1} \pmod{F_a}.$$

This completes the induction. ■

(b) **Claim:** $F_a \mid F_b$ whenever $b \in \mathbb{N}$ is a multiple of a.

Write $b = ma$ for some $m \in \mathbb{N}$. We induct on m.

Base case ($m = 0$): $F_0 = 0$ and $F_a \mid 0$. ✓

Inductive step. Assume $F_a \mid F_{ma}$; we show $F_a \mid F_{(m+1)a}$.

Applying part (a) with $n = ma$:

$$F_{ma+a} \equiv F_{ma} \cdot F_{a+1} \pmod{F_a}.$$

By the inductive hypothesis $F_a \mid F_{ma}$, so $F_{ma} \equiv 0 \pmod{F_a}$. Therefore:

$$F_{(m+1)a} \equiv 0 \cdot F_{a+1} = 0 \pmod{F_a},$$

i.e., $F_a \mid F_{(m+1)a}$. ■

Problem 2: Computing Modular Inverses

(a) **Using the Pulverizer.** We run the extended Euclidean algorithm on 43 and 15:

$$\begin{aligned} 43 &= 2 \cdot 15 + 13 \\ 15 &= 1 \cdot 13 + 2 \\ 13 &= 6 \cdot 2 + 1 \\ 2 &= 2 \cdot 1 + 0 \end{aligned}$$

Back-substituting to express 1 as a linear combination of 43 and 15:

$$\begin{aligned} 1 &= 13 - 6 \cdot 2 \\ &= 13 - 6(15 - 1 \cdot 13) = 7 \cdot 13 - 6 \cdot 15 \\ &= 7(43 - 2 \cdot 15) - 6 \cdot 15 = 7 \cdot 43 - 20 \cdot 15. \end{aligned}$$

Thus $-20 \cdot 15 \equiv 1 \pmod{43}$. Converting to $[0, 43)$: $-20 \equiv 23 \pmod{43}$.

Verification: $15 \cdot 23 = 345 = 8 \cdot 43 + 1 \equiv 1 \pmod{43}$. ✓

The inverse of $15 \pmod{43}$ is $\boxed{23}$.

(b) Using Fermat's Little Theorem. Since 43 is prime and $\gcd(15, 43) = 1$, Fermat's Little Theorem gives $15^{42} \equiv 1 \pmod{43}$, so $15^{-1} \equiv 15^{41} \pmod{43}$.

We compute 15^{41} by repeated squaring:

$$\begin{aligned} 15^1 &\equiv 15 \pmod{43} \\ 15^2 &\equiv 225 = 5 \cdot 43 + 10 \equiv 10 \pmod{43} \\ 15^4 &\equiv 10^2 = 100 = 2 \cdot 43 + 14 \equiv 14 \pmod{43} \\ 15^8 &\equiv 14^2 = 196 = 4 \cdot 43 + 24 \equiv 24 \pmod{43} \\ 15^{16} &\equiv 24^2 = 576 = 13 \cdot 43 + 17 \equiv 17 \pmod{43} \\ 15^{32} &\equiv 17^2 = 289 = 6 \cdot 43 + 31 \equiv 31 \pmod{43} \end{aligned}$$

Since $41 = 32 + 8 + 1$:

$$15^{41} \equiv 31 \cdot 24 \cdot 15 \pmod{43}.$$

$31 \cdot 24 = 744 = 17 \cdot 43 + 13 \equiv 13 \pmod{43}$, $\quad 13 \cdot 15 = 195 = 4 \cdot 43 + 23 \equiv 23 \pmod{43}$.

The inverse of $15 \pmod{43}$ is $\boxed{23}$.

Problem 3: Primality Testing

(a) Claim: If the Fermat test returns COMPOSITE, then n is indeed composite.

Proof. If the test returns COMPOSITE, then $a^{n-1} \not\equiv 1 \pmod{n}$ for the chosen a. We prove the contrapositive: if n is prime, then $a^{n-1} \equiv 1 \pmod{n}$ for all $1 \leq a \leq n - 1$.

If n is prime and $1 \leq a \leq n - 1$, then $\gcd(a, n) = 1$. By Fermat's Little Theorem, $a^{n-1} \equiv 1 \pmod{n}$. Therefore, if $a^{n-1} \not\equiv 1 \pmod{n}$, then n cannot be prime, i.e., n is composite. ■

(b) Claim: If p is prime and $x^2 \equiv 1 \pmod{p}$, then $x \equiv \pm 1 \pmod{p}$.

Proof. From $x^2 \equiv 1 \pmod{p}$ we get $p \mid x^2 - 1 = (x-1)(x+1)$. Since p is prime, Lemma 9.4.2 (if a prime divides a product, it divides one of the factors) implies $p \mid (x - 1)$ or $p \mid (x + 1)$. Hence $x \equiv 1 \pmod{p}$ or $x \equiv -1 \pmod{p}$. ■

(c) Claim: If the Miller–Rabin test returns COMPOSITE, then n is indeed composite.

Proof. We prove the contrapositive: *if n is prime, the algorithm never returns* COMPOSITE.

Assume n is prime and $1 \leq a \leq n - 1$. Since n is prime, $\gcd(a, n) = 1$, so by Fermat's Little Theorem $a^{n-1} \equiv 1 \pmod{n}$, i.e., $x_e \equiv 1 \pmod{n}$. This means the first condition for returning COMPOSITE ($x_e \not\equiv 1$) is *never* triggered.

For the second condition: suppose for some i that $x_{i+1} \equiv 1 \pmod{n}$. Then $x_i^2 = x_{i+1} \equiv 1 \pmod{n}$. Since n is prime, part (b) guarantees that the only square roots of 1 mod n are ± 1, so $x_i \equiv \pm 1 \pmod{n}$. Thus the pair (x_i, x_{i+1}) with $x_{i+1} \equiv 1$ but $x_i \not\equiv \pm 1$ *never occurs.*

Hence, if n is prime, the algorithm always returns I DON'T KNOW. Contrapositive: if the algorithm returns COMPOSITE, then n is composite. ■

Problem 4: Pulverizer State Machine

(a) We verify that **(Pres1)**, **(Pres2)**, and **(Pres3)** are each preserved across one transition.

Consider a state (x, y, s, t, u, v) satisfying all three predicates, with $y > 0$. Let $q = x \operatorname{div} y$ and $r = x \operatorname{rem} y$, so $x = qy + r$. The successor state is $(x', y', s', t', u', v') = (y,\ r,\ u,\ v,\ s - qu,\ t - qv)$.

(Pres1):

$$\gcd(x', y') = \gcd(y, r) = \gcd(y, x - qy) = \gcd(x, y) = \gcd(a, b). \checkmark$$

(Pres2):

$$s'a + t'b = ua + vb = y = x'. \checkmark$$

(The last two equalities use (Pres3) and the definition of x' in the old state.)

(Pres3):

$$u'a + v'b = (s - qu)a + (t - qv)b = (sa + tb) - q(ua + vb) = x - qy = r = y'. \checkmark$$

(Using (Pres2) and (Pres3) in the old state.)

Since all three predicates are preserved across every transition, they form a preserved invariant. ■

(b) **Correctness upon termination.** At termination the machine halts because $y = 0$ (transitions only fire when $y > 0$). At this point:

- By **(Pres1)**: $\gcd(x, 0) = x = \gcd(a, b)$.
- By **(Pres2)**: $sa + tb = x = \gcd(a, b)$.

Hence the output coefficients s, t satisfy Bézout's identity $\gcd(a, b) = sa + tb$, so the machine computes a correct answer. ■

(c) The x-and-y components of the Pulverizer state evolve by exactly the same rule $(x, y) \to (y,\ x \bmod y)$ as the Euclidean algorithm, so the machine performs precisely the same sequence of divisions and terminates after exactly the same number of transitions. ■